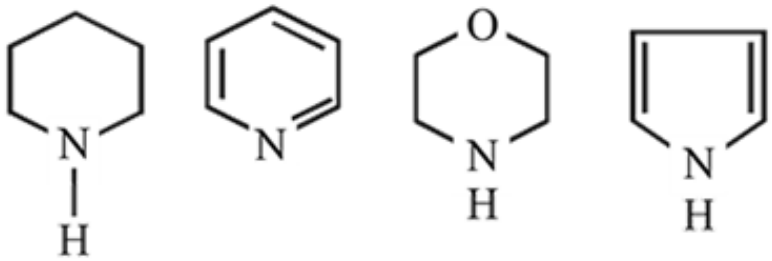
# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

The order of basicity of the compounds



(I) (II) (III) (IV)

is

**(A)** $\mathrm{IV} > \mathrm{I} > \mathrm{III} > \mathrm{II}$

**(B)** $\mathrm{I} > \mathrm{III} > \mathrm{II} > \mathrm{IV}$

**(C)** $\mathrm{III} > \mathrm{I} > \mathrm{IV} > \mathrm{II}$

**(D)** $\mathrm{II} > \mathrm{I} > \mathrm{III} > \mathrm{IV}$

**Q.2**

Identify the prdouct C in the series

$$\mathrm{CH_3CN} \xrightarrow{\mathrm{Na/C_2H_5OH}} \mathrm{A} \xrightarrow{\mathrm{HNO_2}} \mathrm{B} \xrightarrow{\mathrm{Cu/573K}} \mathrm{C}$$

**(A)** $\mathrm{CH_3COOH}$

**(B)** $\mathrm{CH_3CH_2NHOH}$

**(C)** $\mathrm{CH_3CONH_2}$

**(D)** $\mathrm{CH_3CHO}$

**Q.3**

In acid medium nitrobenzene is reduced to aniline as shown in the reaction

$$\mathrm{C_6H_5 - NO_2 + 6[H] \rightarrow C_6H_5 - NH_2 + 2H_2O}$$

The reducing agent used in this reaction is .....

**(A)** $\mathrm{LiAlH_4}$

**(B)** $\mathrm{Sn/HCl}$

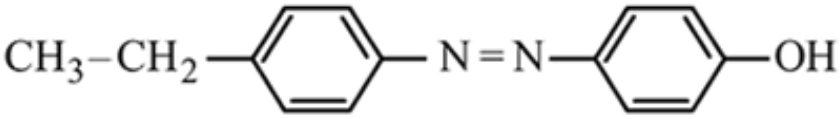
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**(C)** Na/alcohol

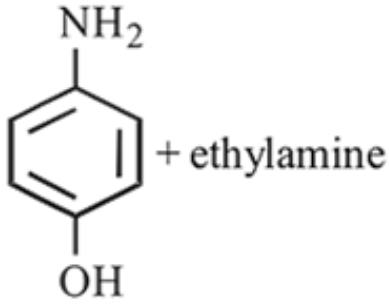
**(D)** $H_2/Ni$

**Q.4**

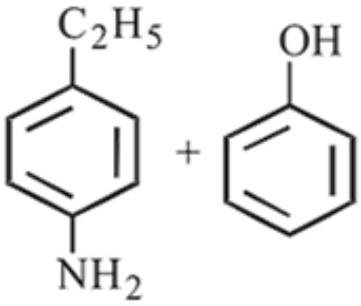
The starting reagents needed to make the azo compound shown below



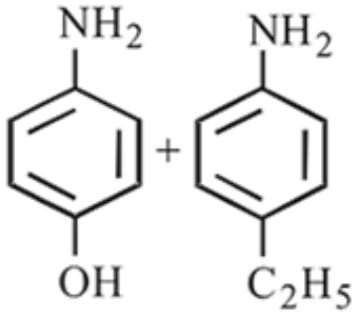
**(A)**



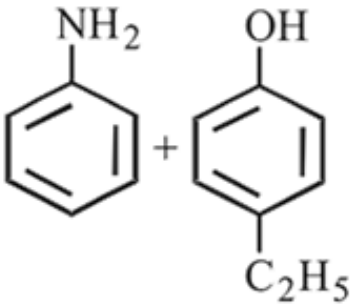
**(B)**



**(C)**



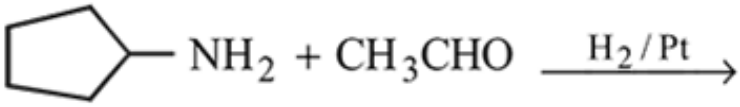
**(D)**



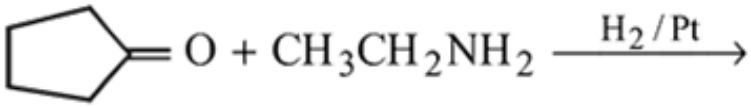
**Q.5**

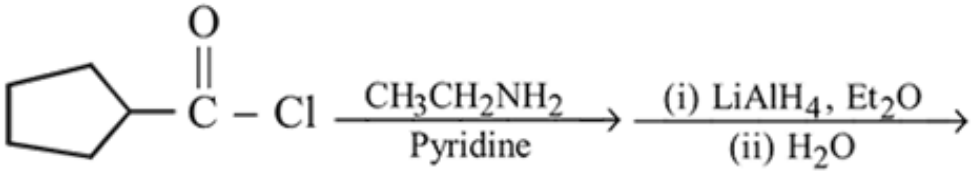
Which of the following does not give N-ethyl cyclopentylamine as major product?

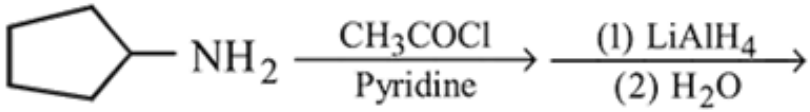
**(A)**



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**(B)** 

**(C)** 

**(D)** 

**Q.6**

Match the compounds in List I with their nature from List II, as seen in aqueous medium

| List I | List II |
|---|---|
| I. Acetamide | A. Acidic |
| II. Benzonitrile | B. Basic |
| III. Triethylamine | C. Neutral |
| IV. Phenol | |

**(A)** I-C; II-C; III-B; IV-A

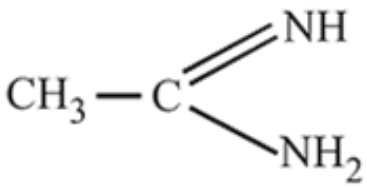
**(B)** I-B; II-C; III-C; IV-A

**(C)** I-C; II-B; III-B; IV-C

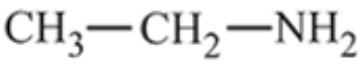
**(D)** I-A;II-A; III-C;IV-B

**Q.7**

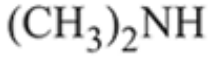
The correct order of basicities of the following compounds is



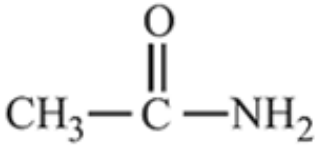
1



2



3



4

**(A)** $2 > 1 > 3 > 4$

**(B)** $1 > 3 > 2 > 4$

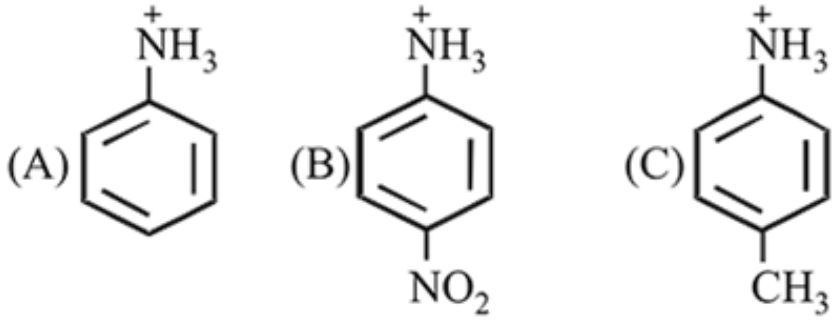
**(C)** $3 > 1 > 2 > 4$

**(D)** $1 > 2 > 3 > 4$

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**Q.8**

Arrange the following three compounds in order of decreasing acidity




**(A)** $\mathrm{B} > \mathrm{A} > \mathrm{C}$

**(B)** $\mathrm{B} > \mathrm{C} > \mathrm{A}$

**(C)** $\mathrm{C} > \mathrm{B} > \mathrm{A}$

**(D)** $C > A > B$

**Q.9**

The reduction of which of the following compound would yield secondary amine?

**(A)** Alkyl nitrite

**(B)** Carbylamine

**(C)** Primary amine

**(D)** Secondary nitro compound

**Q.10**

Which of the following leads to carbon-carbon double bond?

**(A)** $1^{\circ}$ Amine $+\mathrm{RCHO} \rightarrow$

**(B)** $2^{\circ}$ Amine $+\mathrm{R_2CO} \rightarrow$

**(C)** $2^{\circ}$ Amine $+\mathrm{RCHO} \rightarrow$

**(D)** Both (b) and (c)

**Q.11**

In Hofmann bromamide degradation, one of the important steps is the migration of

**(A)** an alkyl group without its electron pair to electron deficient $\mathrm{N}$ atom.

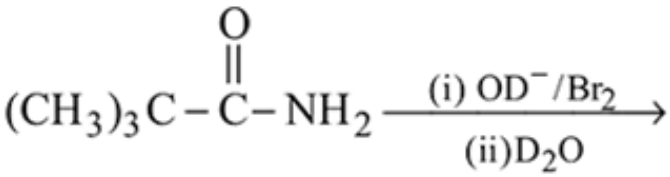
**(B)** an alkyl group with its electron pair to electron deficient $\mathrm{O}$ atom.

**(C)** an alkyl group with its electron pair to electron rich $\mathrm{N}$ atom.

**(D)** an alkyl group with its electron pair to electron deficient $\mathrm{N}$ atom

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**Q.12**



Product P is

**(A)** $(CH_3)_3 CNH_2$

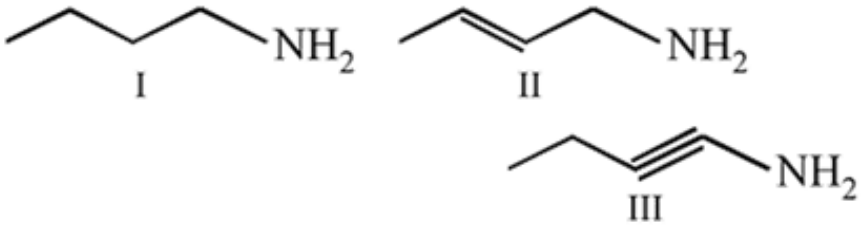
**(B)** $(CH_3)_3 CNHD$

**(C)** $(CH_3)_3 CND_2$

**(D)** no reaction

**Q.13**

The correct order of decreasing basic character of the three aliphatic primary amines is



**(A)** $I > II > III$

**(B)** $III > II > I$

**(C)** $I > II \approx III$

**(D)** $I = II \equiv III$

**Q.14**

Introduction of a methyl group in ammonia markedly increases the basic strength of ammonia in aq. solution, introduction of the second methyl group increases only marginally the basic strength of methyl amine in water. This is due to

**(A)** different type of hybridisation in the two amines.

**(B)** protonated dimethyl amine is more solvated than methyl amine.

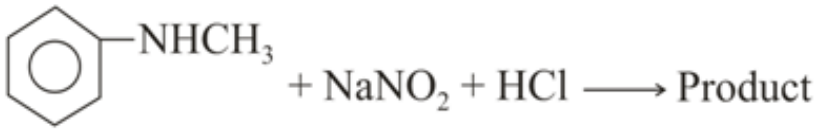
**(C)** protonated dimethyl amine is more solvated than the protonated methyl amine.

**(D)** protonated dimethyl amine is less stable than the protonated methyl amine.

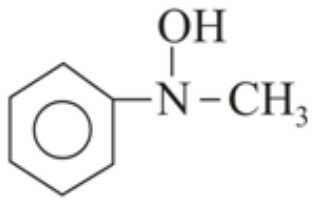
**Q.15**

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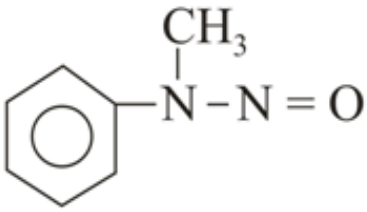
Predict the product :



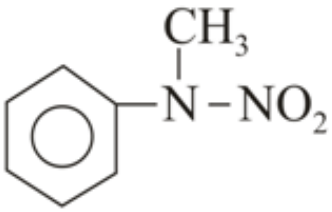
**(A)**



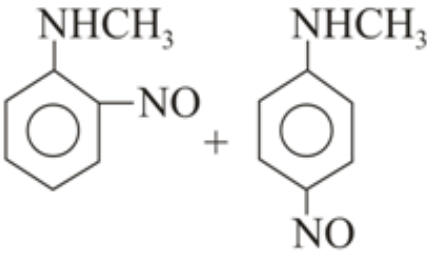
**(B)**



**(C)**



**(D)**



## Answers & Solutions

**Q.1 Answer:**

I > III > II > IV

**Solution:**

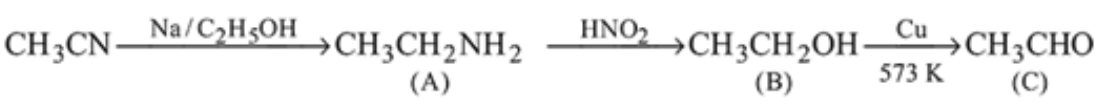
**The order of basicity is** I > III > II > IV

**The lone pair of electrons on** N **is more readily available for protonation in I and III then in II. III contains an oxygen atom which has - I effect due to which it is less basic than I. In compound IV lone pair of e** $^{-}s$ **on** $N$ **-atom is contributed towards the aromatic sextet formation and hence is not at all available for protonation. Hence option (b) is correct.**

**Q.2 Answer:**

$CH_3CHO$

**Solution:**



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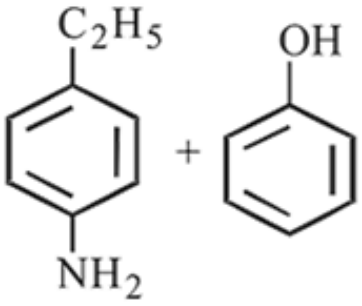
**Q.3 Answer:**

$Sn/HCl$

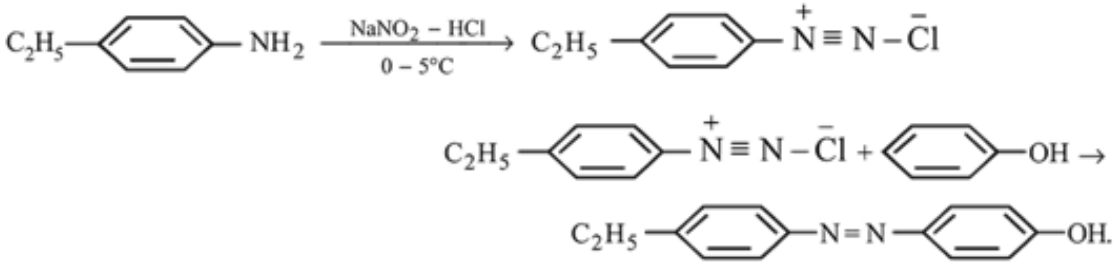
**Solution:**

$$C_6H_5NO_2 + 6H \xrightarrow{Sn+HCl} C_6H_5 - NH_2 + 2H_2O$$

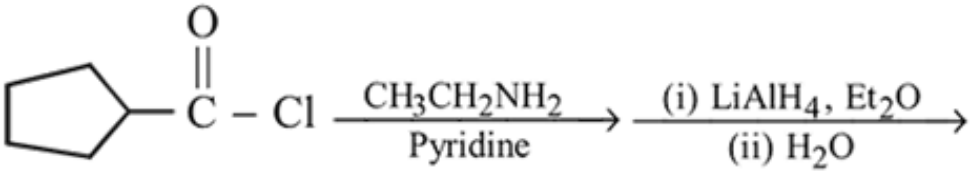
**Q.4 Answer:**



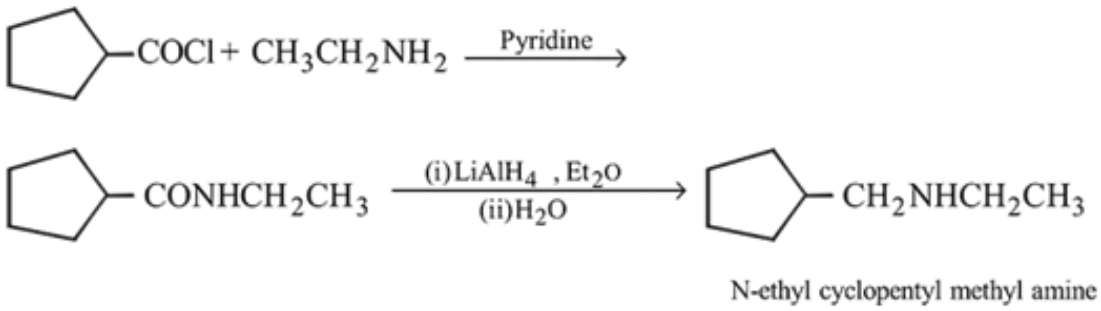
**Solution:**



**Q.5 Answer:**



**Solution:**



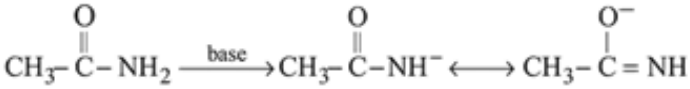

**Q.6 Answer:**

I-C; II-C; III-B; IV-A

**Solution:**

**Acetamide is basic due to the presence of lone pair of electrons of $N$; it is also acidic because its conjugate base shows resonance.**

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**(more stable due to -ve charge on $O$ )**

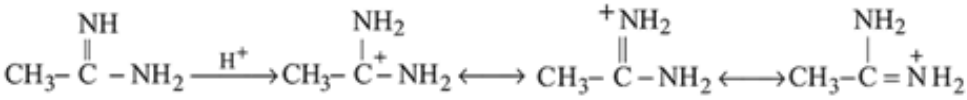
**Benzonitrile $\left(C_6H_5C=N\right)$ acts as an electrophile (Lewis acid) due to electron deficiency of $C$ of $CN$ as well as nucleophile (Lewis base) in nature due to presence of lone pair electrons on $N$; hence it is neutral Triethylamine and phenol are basic and acidic in nature respectively.**

**Q.7 Answer:**

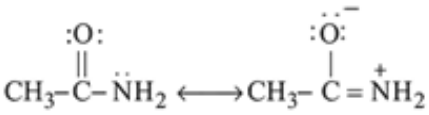
$1>3>2>4$

**Solution:**

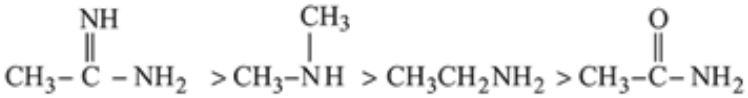
**Guanidine is most basic because its conjugate acid is stabilized by two equivalent resonance structures,**



**Further $2^{\circ}$ amines are more basic than $1^{\circ}$, while amides are least basic due to delocalisation of the lone pair of electrons on $N$**



**Thus the correct order of basicities is**



**Q.8 Answer:**

$B>A>C$

**Solution:**

**The electrons withdrawing groups decrease the basic character and increase the acid character by inductive effect and resonance.**

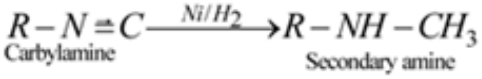
**Q.9 Answer:**

Carbylamine

**Solution:**

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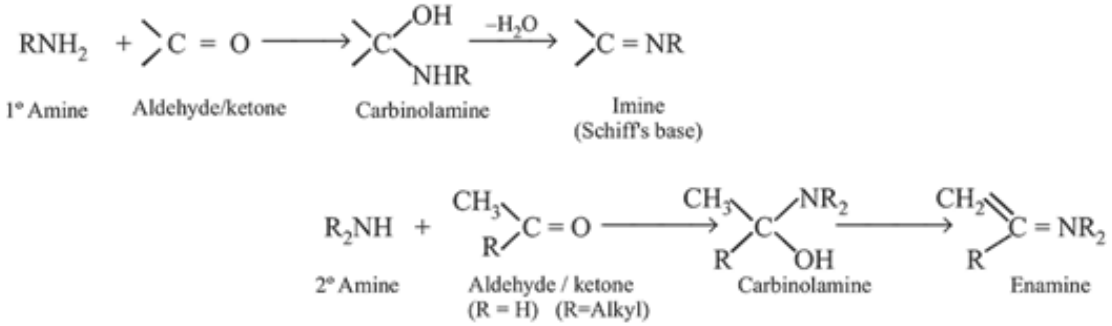
**Carbylamine (or isocyanides) give secondary amine on reduction**




**Q.10 Answer:**

Both (b) and (c)

**Solution:**

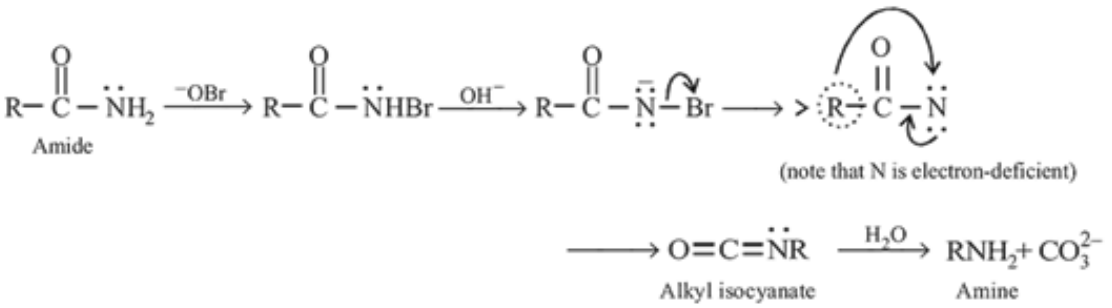



**Q.11 Answer:**

an alkyl group with its electron pair to electron deficient $\mathrm{N}$ atom

**Solution:**

**For this one should remember that Hofmann's degradation involves migration to electron deficient nitrogen atom, hence the alkyl group will migrate with its bonding pair of electrons.**

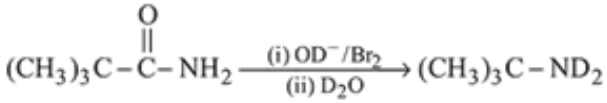



**Q.12 Answer:**

$(CH_3)_3 CND_2$

**Solution:**

**Remember that in Hofmann rearrangement, the two original $\mathrm{H}$ atoms of the $-CONH_2$ group are removed by base $(OH^-)$ and new H's are derived from $H_2O$.**



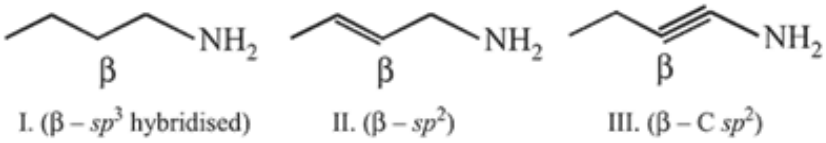

**Q.13 Answer:**

$\mathrm{I} > \mathrm{II} > \mathrm{III}$

**Solution:**

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**Note the point of difference in the given compounds which here lies at $\beta$ -carbon. In I, II, III, the $\beta$ -carbon atoms are $sp^3$, $sp^2$ and $sp$ hybridised respectively which in turn cause the difference in their $s$ -character. We know that more is the $s$ character of an atom, greater will be its electron-withdrawing nature. Thus $sp$ $(50\% s$ character) hybridised carbon is most electron-withdrawing, while $sp^3(25\%$ $s$ -character) is least electron-withdrawing. Further, we know that presence of an electron-withdrawing group decreases basicity of an amine. Thus**



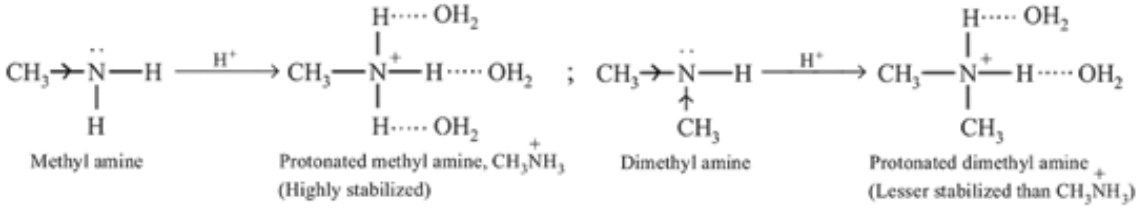

I. (β – $sp^3$ hybridised) II. (β – $sp^2$) III. (β – C $sp^2$)

**Q.14 Answer:**

protonated dimethyl amine is less stable than the protonated methyl amine.

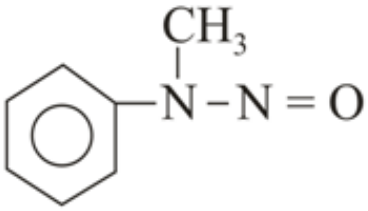
**Solution:**

**The basic character of an amine in water is determined by (i) electron availability on the N atom and (ii) the extent of stabilization of the cation (protonated amine) due to solvation by hydrogen bonding**



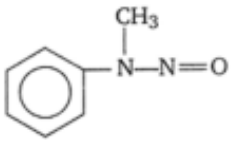

Methyl amine — Protonated methyl amine, $CH_3\overset{+}{N}H_3$ (Highly stabilized) — Dimethyl amine — Protonated dimethyl amine (Lesser stabilized than $CH_3\overset{+}{N}H_3$)

**Q.15 Answer:**

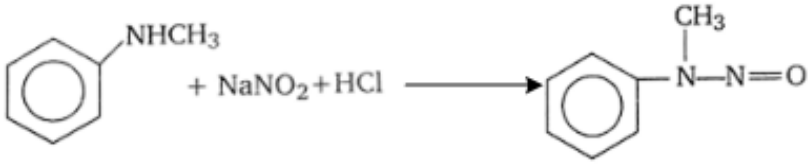



**Solution:**

.



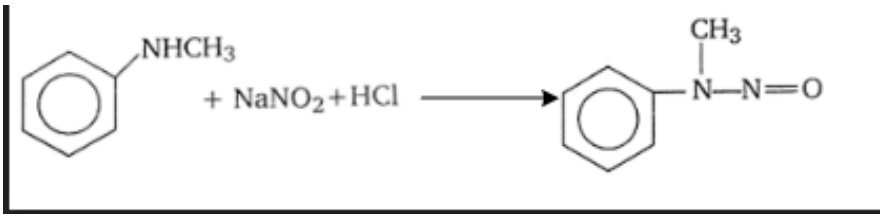

**With nitric acid $(NaNO_2) + HCl)$, primary aromatic amines give azo compounds and secondary aromatic amines gives nitroso compound**




**With nitric acid $(NaNO_2) + HCl)$, primary aromatic amines give azo compounds and secondary**

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**aromatic amines gives nitroso compound**




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